

Shavait Pandita

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (Civil Engineering)	Indian Institute of Technology, Indore	7.78 (Current)	2025 – Present
Senior Secondary	JKBOSE	89.8%	2025
Secondary	JKBOSE	96.6%	2023

Research

- **AAAI-27 Undergraduate Consortium – Research Proposal** *2026*
  Uncertainty-Aware Graph Intelligence for Autonomous eVTOL Traffic
  - Proposed a GNN predictor for conflicts, sector congestion and vertiport queues, with calibrated uncertainty
  - Defined the Airspace Risk Horizon lead-time metric and uncertainty-driven selective autonomy with a safety shield
  - Planned BlueSky simulation on a DFW-style network (50–200 aircraft) and an open-source benchmark

Projects

- **FlashGuard – Real-Time Fraud Detection on Pathway (IITISoC 2026)** *Jul 2026*
  Pathway, XGBoost, Half-Space Trees, LangGraph, RAG, Redis, Docker
  - Built the core streaming pipeline in Pathway: Kafka ingestion, live joins with profiles and sanctions watchlists, and 10-min sliding-window analytics per account
  - Ran an XGBoost + Half-Space Trees ensemble as a Pathway UDF, scoring and learning online; clear cases decided in ∼15–20 ms
  - Built a live compliance RAG on Pathway's DocumentStore that hot-reloads AML/KYC policies without a restart
  - Used Pathway event streams as an audit ledger of agent decisions; ambiguous cases go to LangGraph, with a Redis watchdog cutting token use 80%+
- **Tacotron2 + HiFi-GAN – End-to-End Neural Text-to-Speech** *2026*
  PyTorch, Tacotron2, HiFi-GAN, HF Accelerate, LJSpeech
  - Built a modular TTS stack in PyTorch (text frontend → Tacotron2 → HiFi-GAN) with a decoupled tokenizer, so new scripts (e.g., Indic) need only vocabulary/frontend changes
  - Implemented Tacotron2 from scratch: Conv1D + BiLSTM encoder, location-sensitive attention for robust text–audio alignment, 2-layer LSTM decoder with PreNet, 5-layer Conv1D PostNet
  - Streaming-ready inference: step-wise autoregressive mel decoding with stop-token gating into a fully convolutional HiFi-GAN (256× upsampling, MRF, MPD + MSD)
  - Curated LJSpeech mels (dB-scaled, $[-4, 4]$) and generated a synthetic Tacotron2 mel corpus to fine-tune HiFi-GAN via Accelerate; UTMOS 3.37 → 3.74 (Griffin-Lim: 1.88)
- **GPT from Scratch – Decoder-Only Transformer** *2026*
  PyTorch, Transformers, RoPE, Tiny Shakespeare
  - Implemented a ∼10M parameter decoder-only Transformer from scratch in PyTorch (6 layers, 6 heads, 384-d): causal multi-head attention, RoPE, GELU, residual connections
  - Built a modular data, training and generation pipeline with a character-level tokenizer, 90/10 split, validation tracking, periodic + best-model checkpointing, and gradient clipping
  - Added weight tying, label smoothing, and 200-step warmup + cosine LR decay for training stability
  - Added autoregressive generation with temperature and top-k sampling; validation loss dropped from 4.19 to ∼1.95

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL
- **AI/ML:** PyTorch, XGBoost, Pandas, NumPy, Transformers, Sequence Modeling, Fine-tuning (LoRA), NLP
- **Speech & Audio:** TTS, Mel-Spectrograms, Neural Vocoders, Streaming Inference
- **Data Engineering:** Dataset Curation, Synthetic Data Generation, Streaming Pipelines, Feature Engineering
- **GenAI:** LLMs, RAG, Embeddings, Vector Databases, LangGraph
- **Backend & Web:** Node.js, Express.js, React, REST APIs, Firebase, Razorpay, PostgreSQL
- **System Design:** Scalability, Caching, Kafka, Redis, Pathway
- **Tools:** Git, GitHub, Docker

Positions of Responsibility

- **Web Development Associate,** E-Summit 2026, IIT Indore *May 2026 – Aug 2026*
- **Volunteer,** Programming Club, IIT Indore *2025 – Present*
- **Volunteer,** AI and Data Science, Maths Club, IIT Indore *2025 – Present*

Achievements

- NSEP (National Standard Examination in Physics) – Centre Top 1% *2024-25*
- Qualified for Regional Mathematics Olympiad (RMO) *2024-25*
- IISER Aptitude Test (IAT) – AIR 1389 *2025*
- Codeforces – Maximum rating 1315; best rank 2683 among 32,000 participants *2025 – Present*
- Data Structures & Algorithms – Solved 250+ problems across LeetCode and Codeforces *2025 – Present*